

COMBINATORICS FINAL EXAM

ROB TOMPKINS

1. (Multinomial Coefficients) A committee of 12 people is to be divided into four subcommittees of sizes 3, 4, 2, and 3.

- (a) How many ways can this be done if the four subcommittees have distinct responsibilities?
- (b) How many ways can this be done if two of the subcommittees of size 3 have identical responsibilities, while the others are distinct?
- (c) Explain why the multinomial coefficient

$$\binom{12}{3.4.2.3}$$

appears naturally in part (a).

Solution. (a) Label the responsibilities A, B, C, D . Choosing their members successively gives

$$\binom{12}{3} \binom{9}{4} \binom{5}{2} \binom{3}{3} = \frac{12!}{3!4!2!3!} = 277\,200.$$

(b) Interchanging the two size-3 blocks does not change the assignment. Each assignment was counted exactly $2!$ times, so the answer is $277\,200/2 = 138\,600$.

(c) The multinomial coefficient $\binom{12}{3,4,2,3} = 12!/(3!4!2!3!)$ counts allocations to four labeled blocks of those sizes. Equivalently, order all twelve people and cut the order into blocks of lengths 3, 4, 2, 3; dividing by the factorial of each block size forgets its internal order. $\square$

2. (Binomial Identities) Prove the identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

Give **two different proofs**

- (a) An algebraic proof using the binomial theorem.
- (b) A combinatorial proof by counting the same set in two ways.

Proof. Correction: The right-hand side must be $\binom{2n}{n}$, not $\binom{2n}{k}$, since k is the summation index.

(a) Compare coefficients of x^n in

$$(1+x)^n(1+x)^n = (1+x)^{2n}.$$

On the left the coefficient is $\sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \sum_{k=0}^n \binom{n}{k}^2$; on the right it is $\binom{2n}{n}$.

(b) Choose n people from two disjoint groups of n people each. Directly there are $\binom{2n}{n}$ choices. If exactly k come from the first group, there are $\binom{n}{k} \binom{n}{n-k} = \binom{n}{k}^2$ choices. Summing over the mutually exclusive possibilities $0 \leq k \leq n$ proves the identity. $\square$

3. (Ramsey Numbers)

(a) Show that if $m \geq 3$ and $n \geq 3$ are integers then

$$R(m, n) \leq R(m-1, n) + R(m, n-1).$$

(b) Show that if $R(m-1, n)$ and $R(m, n-1)$ are both even and greater than 2, then

$$R(m, n) \leq R(m-1, n) + R(m, n-1) - 1.$$

(c) Show that $R(4, 3) = 9$, $R(5, 3) = 14$, and $R(4, 4) = 18$

Proof. Notation corrections: Read $R(m, n-1)$ as $R(m, n-1)$ and $R(5, 3)$ as $R(5, 3)$. We use $R(m, n)$ for the least N such that every red–blue edge coloring of K_N contains a red K_m or a blue K_n .

(a) Put $a = R(m-1, n)$ and $b = R(m, n-1)$. At any vertex of K_{a+b} , either at least a incident edges are red or at least b are blue. In the first case its red neighbors contain a blue K_n or a red K_{m-1} , which extends with the vertex to a red K_m . The second case is symmetric. Thus $R(m, n) \leq a + b$.

(b) Suppose a coloring of K_{a+b-1} avoids both forbidden cliques. The preceding argument implies that every vertex has red degree at most $a-1$ and blue degree at most $b-1$. Their sum is $a+b-2$, so both bounds are equalities. If a, b are even, the red graph has an odd number $a+b-1$ of vertices, all of odd degree $a-1$. Its degree sum is odd, contradicting the handshake lemma. Hence $R(m, n) \leq a + b - 1$.

(c) We give both upper and lower bounds. First, $R(2, t) = t$ and $R(s, t) = R(t, s)$. Also $R(3, 3) = 6$: among the five edges at a vertex of K_6 , three have one color, and either a pair of their endpoints has that color or all three intervening edges have the other color. For the lower bound, color a 5-cycle red and its complement blue; neither graph has a triangle. Part (b), followed by part (a), now gives

$$R(4, 3) \leq 6 + 4 - 1 = 9, \quad R(5, 3) \leq 9 + 5 = 14, \quad R(4, 4) \leq 9 + 9 = 18.$$

Here are explicit colorings establishing the reverse bounds. In each construction, unlisted edges receive the other color.

$R(4, 3) > 8$. On $\mathbb{Z}/8\mathbb{Z}$, color an edge blue when its endpoints differ by ± 1 or 4. This is an 8-cycle with opposite vertices joined. There is no blue triangle: the blue neighbors of 0 are 1, 4, 7, and no two of them are adjacent; translation handles every other vertex. An independent set of size 4 in the 8-cycle must consist of alternating vertices, but either alternating set contains an opposite pair. Thus the blue graph has no independent set of size 4, so the red graph has no K_4 .

$R(5, 3) > 13$. On $\mathbb{Z}/13\mathbb{Z}$, color an edge blue when the difference is ± 1 or ± 5 . The neighbors 1, 5, 8, 12 of 0 have no blue edges between them, so translation again proves triangle-freeness. Suppose five vertices were independent, and list the five positive circular gaps between consecutive chosen vertices. Each gap is at least 2 and their sum is 13. Apart from order, the only possibilities are

$$(5, 2, 2, 2, 2), \quad (4, 3, 2, 2, 2), \quad (3, 3, 3, 2, 2).$$

The first has a forbidden gap 5. In the second, the gap 3 must neighbor a gap 2 on at least one side; in the third, there is also an adjacent 3, 2 pair. Such a pair puts two chosen vertices at circular distance 5, again a blue edge. Thus no independent set of size 5 exists.

$R(4, 4) > 17$. On $\mathbb{Z}/17\mathbb{Z}$, color an edge red when its nonzero difference belongs to

$$Q = \{1, 2, 4, 8, 9, 13, 15, 16\},$$

the set of nonzero squares modulo 17. Since $-1 \in Q$, this defines an undirected graph. Translation and multiplication by a member of Q preserve red edges and take any red edge to $\{0, 1\}$. The common red neighbors of 0 and 1 are 2, 9, 16. Their pairwise differences are 7, 7, 14, none in Q , so this edge cannot belong to a red K_4 . Consequently there is no red K_4 anywhere. Multiplication by the nonsquare 3 interchanges square and nonsquare differences and hence interchanges colors. There is no blue K_4 either. Combining these constructions with the upper bounds gives $R(4, 3) = 9$, $R(5, 3) = 14$, and $R(4, 4) = 18$. $\square$

4. (Stirling Numbers of the Second Kind) Let $S(n, k)$ denote the Stirling numbers of the second kind, the number of ways to partition an n -element set into k nonempty unlabeled blocks.

(a) Prove the recurrence

$$S(n, k) = kS(n-1, k) + S(n-1, k-1).$$

(b) Use the recurrence to compute $S(5, 2)$, $S(5, 3)$, and $S(6, 3)$.

(c) Give a direct combinatorial interpretation of $S(6, 3)$.

Proof. (a) Distinguish one element, say n . If it is alone in its block, partition the other elements into $k-1$ blocks, giving $S(n-1, k-1)$ choices. Otherwise first partition the other elements into k blocks and choose one of the k blocks to receive n , giving $kS(n-1, k)$. The

cases are disjoint and exhaustive. The boundary values are $S(0,0) = 1$, $S(n,0) = 0$ for $n > 0$, and $S(n,k) = 0$ for $k > n$.

(b) Starting with $S(n,1) = S(n,n) = 1$, the recurrence gives

$$S(3,2) = 3, \quad S(4,2) = 2 \cdot 3 + 1 = 7, \quad S(4,3) = 3 \cdot 1 + 3 = 6.$$

Hence

$$S(5,2) = 15, \quad S(5,3) = 25, \quad S(6,3) = 3 \cdot 25 + 15 = 90.$$

(c) $S(6,3)$ counts divisions of six distinct objects into three nonempty, unlabeled groups. Their possible block sizes are $(4,1,1)$, $(3,2,1)$, and $(2,2,2)$, yielding

$$\binom{6}{4} + \binom{6}{3} \binom{3}{2} + \frac{6!}{(2!)^3 3!} = 15 + 60 + 15 = 90.$$

□

5. (Pólya Enumeration) Consider coloring of the vertices of a square using three colours: red, blue, and green. Two colourings are considered equivalent if one can be obtained from the other by a rotation or reflection of the square.

- Identify the symmetry group acting on the vertices of the square.
- Write the cycle types of all symmetries of the square.
- Use Burnside's Lemma or Pólya's Enumeration Theorem to determine the number of distinct colourings.
- Briefly explain why simply computing 3^4 over counts the number of essentially different colourings.

Solution. (a) The group is the dihedral group D_4 of order 8 (also denoted D_8), acting on the four vertices.

(b) The cycle types and fixed-coloring counts are

Symmetry	Number	Cycle type	Fixed colorings
Identity	1	1^4	3^4
$90^\circ, 270^\circ$ rotations	2	4	3
180° rotation	1	2^2	3^2
Reflections in diagonals	2	$1^2 2$	3^3
Reflections through side midpoints	2	2^2	3^2

A coloring is fixed exactly when it is constant on each cycle.

(c) Burnside's lemma therefore gives

$$\frac{3^4 + 2 \cdot 3 + 3^2 + 2 \cdot 3^3 + 2 \cdot 3^2}{8} = \frac{168}{8} = 21.$$

Equivalently, the cycle index is $\frac{1}{8}(a_1^4 + 2a_4 + 3a_2^2 + 2a_1^2 a_2)$.

(d) The number 3^4 counts colorings of labeled vertices, including rotated and reflected copies separately. Orbit sizes vary because some colorings have symmetries, so simply dividing 3^4 by 8 would also be incorrect. $\square$

6. (Rubik's Cube Combinatorics)

- (a) A standard $3 \times 3 \times 3$ Rubik's Cube has 8 corner cubies and 12 edge cubies. Ignoring the physical how many ways can the 8 corner cubies be permuted and oriented, and how many ways can the 12 edge cubies can be permuted and oriented? Find the total number of reachable $3 \times 3 \times 3$ Rubik's Cube positions.
- (b) A standard $5 \times 5 \times 5$ Rubik's Cube, also called the **Professor's Cube**, as several types of movable pieces: corners, edges, wing edges, and center pieces. Find the total number of reachable $5 \times 5 \times 5$ Rubik's Cube Positions

Solution. We count ordinary color patterns relative to the six fixed face centers; rotations of the whole cube give no extra factor.

(a) Ignoring physical constraints, the corner count is $8!3^8$ and the edge count is $12!2^{12}$, since each corner has three orientations and each edge has two. Legal moves impose three constraints: the total corner twist is 0 modulo 3, the total edge flip is 0 modulo 2, and the corner and edge permutations have the same parity. These conditions are also sufficient for reachability. Thus

$$N_3 = \frac{8!3^8 12!2^{12}}{3 \cdot 2 \cdot 2} = 8!3^7 12!2^{10} = 43\,252\,003\,274\,489\,856\,000.$$

(b) There are 8 corners, 12 middle edges, 24 wings, and two separate orbits of 24 movable centers each (diagonal centers and axial centers). Corners and middle edges contribute N_3 . Wings contribute $24!$: their orientation is determined by their position. Each center orbit contains four indistinguishable pieces of each of six colors, contributing $24!/(4!)^6$. Parity restrictions on labeled centers impose no further reduction on color patterns, since swapping equal-colored centers changes the labeled parity without changing the pattern. The cube's standard reachability conditions therefore give

$$N_5 = N_3 24! \left(\frac{24!}{(4!)^6} \right)^2 = \frac{8! 12! 3^7 2^{10} (24!)^3}{(4!)^{12}} \approx 2.8287094228 \times 10^{74}.$$

This assumes unmarked centers, as on a standard Professor's Cube. A reference for these piece-orbit and reachability constraints is Jaap Scherphuis, *Professor's Cube*,

<https://www.jaapsch.net/puzzles/cube5.htm>.

$\square$

7. (Combinatorics of Continued Fractions) A finite simple continued fraction has the form

$$[a_1, a_2, \dots, a_k] = a_1 + \frac{1}{a_2 + \frac{1}{\dots + \frac{1}{a_k}}},$$

where $a_i \in \mathbb{Z}^+$.

Let $C(n)$ be the number of ordered compositions of n , that is the number of ways to write

$$n = a_1 + a_2 + \dots + a_k,$$

where $k \geq 1$ and each $a_i \geq 1$.

- (a) Find the simple continued fraction expansion of $\frac{415}{93}$. Then verify your answer by converting the continued fraction back into a rational number.
- (b) Find the five partial quotients in the simple continued fraction of $\sqrt{7}$. That is, write $\sqrt{7} = [a_0; a_1, a_2, a_3, a_4, \dots]$.
- (c) Convert the periodic continued fraction $x = [2; \overline{1, 3}]$ into a quadratic irrational number in simplified radical form. That is, solve for x exactly.
- (d) Let T_n be the number of ways to tile a $1 \times n$ board using squares of size 1×1 and dominoes of size 1×2 .
 - (a) Prove that $T_n = T_{n-1} + T_{n-2}$, with initial values $T_0 = 1$ and $T_1 = 1$.
 - (b) Compute T_2, T_3, T_4, T_5 , and T_6 .
 - (c) Explain why this recurrence is related to the continued fraction

$$1 + \frac{1}{1 + \frac{1}{1 + \dots}}.$$

- (d) Show that the infinite continued fraction

$$x = 1 + \frac{1}{1 + \frac{1}{1 + \dots}}$$

satisfies $x = 1 + \frac{1}{x}$, and conclude $x = \frac{1+\sqrt{5}}{2}$.

Thus, the same Fibonacci-type recurrence appears in both the tiling problem and the convergents of the continued fraction.

Solution. (a) The Euclidean algorithm gives

$$415 = 4 \cdot 93 + 43, \quad 93 = 2 \cdot 43 + 7, \quad 43 = 6 \cdot 7 + 1, \quad 7 = 7 \cdot 1.$$

Therefore $415/93 = [4; 2, 6, 7]$. Reversing the calculation,

$$6 + \frac{1}{7} = \frac{43}{7}, \quad 2 + \frac{7}{43} = \frac{93}{43}, \quad 4 + \frac{43}{93} = \frac{415}{93}.$$

(b) Successive complete quotients after $a_0 = 2$ are

$$\frac{1}{\sqrt{7}-2} = \frac{\sqrt{7}+2}{3}, \quad \frac{\sqrt{7}+1}{2}, \quad \frac{\sqrt{7}+1}{3}, \quad \sqrt{7}+2.$$

Their integer parts are 1, 1, 1, 4; the next complete quotient repeats the first. Thus

$$\sqrt{7} = [2; \overline{1, 1, 1, 4}], \quad (a_0, a_1, a_2, a_3, a_4) = (2, 1, 1, 1, 4).$$

(c) Put $y = [1; 3, 1, 3, \dots]$. Periodicity gives

$$y = 1 + \frac{1}{3 + 1/y}, \quad 3y^2 - 3y - 1 = 0.$$

As $y > 0$, $y = (3 + \sqrt{21})/6$. Consequently

$$x = 2 + \frac{1}{y} = 2 + \frac{\sqrt{21} - 3}{2} = \boxed{\frac{1 + \sqrt{21}}{2}}.$$

(d)(a) For $n \geq 2$, the first tile is either a square, leaving a board of length $n - 1$, or a domino, leaving length $n - 2$. These disjoint cases give $T_n = T_{n-1} + T_{n-2}$. There is one empty tiling and one tiling of a length-1 board, so $T_0 = T_1 = 1$.

(d)(b) The next values are

$$T_2 = 2, \quad T_3 = 3, \quad T_4 = 5, \quad T_5 = 8, \quad T_6 = 13.$$

(d)(c) With $F_0 = 0, F_1 = 1$, we have $T_n = F_{n+1}$. The continued fraction with r entries all equal to 1 is F_{r+1}/F_r , for $r \geq 1$: adjoining an initial 1 replaces p/q by $(p+q)/p$. Thus its numerators and denominators obey the same Fibonacci recurrence.

(d)(d) To justify the limit, let $\phi = (1 + \sqrt{5})/2$ and $\psi = (1 - \sqrt{5})/2$. The formula $F_r = (\phi^r - \psi^r)/\sqrt{5}$ follows from the recurrence and initial values. Since $|\psi/\phi| < 1$, $F_{r+1}/F_r \rightarrow \phi$. Removing the first entry of the infinite continued fraction leaves the same tail, so its positive value satisfies $x = 1 + 1/x$, or $x^2 - x - 1 = 0$. Hence $x = (1 + \sqrt{5})/2$.

The introductory composition count: Although no subpart asks for $C(n)$, it is 2^{n-1} for $n \geq 1$: choose which of the $n - 1$ gaps in a row of n ones receive separators. This counts compositions, not distinct rational values of continued fractions, which can have two finite representations. $\square$

8. (Catalan Numbers) The Catalan numbers are given by

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

- (a) Give three different combinatorial objects counted by C_n .
- (b) Prove the number of balanced parentheses strings with n pairs of parentheses is C_n .
- (c) Compute C_4 , and list all balanced parenthesis strings with 4 pairs of parenthesis.

(d) Show that the Catalan numbers satisfy the recurrence

$$C_{n+1} = \sum_{k=0}^n C_k C_{n-k}.$$

Solution. (a) Three examples are balanced strings with n pairs of parentheses, Dyck paths with n up-steps and n down-steps that never go below height 0, and rooted ordered full binary trees with n internal vertices.

(b) Interpret an opening parenthesis as an up-step and a closing parenthesis as a down-step. There are $\binom{2n}{n}$ unrestricted strings with n of each. For a bad string, reflect its steps through its first visit to height -1 (interchange opening and closing parentheses in that prefix). The result has $n+1$ up-steps and $n-1$ down-steps. This is a bijection: in a path ending at height 2, reflect the prefix through its first visit to height 1 to recover the bad path. Thus, for $n \geq 1$, the number of good strings is

$$\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1} \binom{2n}{n} = C_n.$$

For $n = 0$, the single empty string gives $C_0 = 1$.

(c) $C_4 = \binom{8}{4}/5 = 14$. The complete list is

$((((()))$ $((() ()))$ $((()) ())$
 $((()) ())$ $((() (()))$ $((() () ())$
 $((() () ())$ $((() ((()))$ $((() () (())$
 $((() ((()))$ $((() (() ())$ $((() (() ())$
 $((() (() ())$ $((() () (())$

(d) A nonempty balanced string has a unique form $(A)B$, where the displayed closing parenthesis matches the first opening parenthesis, and A, B are balanced strings. For $n+1$ pairs, if A has k pairs then B has $n-k$ pairs, giving $C_k C_{n-k}$ choices. Summing gives $C_{n+1} = \sum_{k=0}^n C_k C_{n-k}$. $\square$

9. (Generating Functions and Integer Partitions) Let $p_{\leq 4}(n)$ be the number of partitions of n into at most 4 parts.

- (a) Write the ordinary generating function for $p_{\leq 4}(n)$.
- (b) Explain why this is also the generating function for partitions of n whose largest part is at most 4.
- (c) Use the generating function to compute $p_{\leq 4}(8)$.
- (d) List all partitions of 8 into at most 4 parts.

Solution. (a) Including the empty partition at $n = 0$,

$$\sum_{n \geq 0} p_{\leq 4}(n) x^n = \frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)}.$$

(b) In the product, the factor $(1 - x^j)^{-1} = \sum_{r \geq 0} x^{jr}$ chooses the multiplicity of part j . Hence it counts partitions with largest part at most 4. Transposing a Ferrers diagram gives a bijection between these partitions and partitions with at most 4 parts.

(c) The coefficient of x^8 counts nonnegative solutions of $a + 2b + 3c + 4d = 8$. For each c, d , there are $\lfloor (8 - 3c - 4d)/2 \rfloor + 1$ choices of b , and then a is determined. Therefore

$$p_{\leq 4}(8) = \underbrace{(5 + 3 + 2)}_{d=0} + \underbrace{(3 + 1)}_{d=1} + \underbrace{1}_{d=2} = 15.$$

(d) Grouping by the number of parts, the partitions are

$$\begin{aligned} 1 &: 8; \\ 2 &: 7 + 1, 6 + 2, 5 + 3, 4 + 4; \\ 3 &: 6 + 1 + 1, 5 + 2 + 1, 4 + 3 + 1, 4 + 2 + 2, 3 + 3 + 2; \\ 4 &: 5 + 1 + 1 + 1, 4 + 2 + 1 + 1, 3 + 3 + 1 + 1, \\ &\quad 3 + 2 + 2 + 1, 2 + 2 + 2 + 2. \end{aligned}$$

□

10. (Derangements) A derangement of n objects is a permutation with no fixed points. Let D_n denote the number of derangements of n objects.

(a) Use the inclusion-exclusion principle to prove that

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

(b) Compute D_5 .

(c) Explain why D_n is the nearest integer to

$$\frac{n!}{e}.$$

Proof. (a) Let A_i be the set of permutations fixing i . For any specified k elements, exactly $(n - k)!$ permutations fix all of them. Inclusion-exclusion therefore gives

$$D_n = \sum_{k=0}^n (-1)^k \binom{n}{k} (n - k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

(b) Substitution yields

$$D_5 = 120 - 120 + 60 - 20 + 5 - 1 = 44.$$

(c) For $n \geq 1$, the strictly alternating tail of $e^{-1} = \sum_{k=0}^{\infty} (-1)^k/k!$ has magnitude strictly less than its first term. Thus

$$\left| D_n - \frac{n!}{e} \right| < \frac{n!}{(n+1)!} = \frac{1}{n+1} \leq \frac{1}{2}.$$

Since D_n is an integer, it is the unique nearest integer. The restriction $n \geq 1$ is necessary: $D_0 = 1$, whereas the nearest integer to $0!/e$ is 0. $\square$

11. (Graph Colouring) A proper colouring of a graph is an assignment of colours to the vertices of the graph such that no two adjacent vertices (Vertices connected by an edge) share the same colour. Let C_n be the cycle graph with n vertices

- (a) Prove that the number of proper colourings of C_n using k available colours is

$$(k-1)^n + (-1)^n(k-1).$$

- (b) Use the formula to find the number of proper colourings of C_6 using 4 colours.

- (c) Explain why the answer changes depending on whether n is even or odd.

Proof. (a) Assume $n \geq 3$ and first $k \geq 2$. Fix the first vertex's color in a path of n vertices, and let B_n count proper path colorings whose last vertex differs from the first. There are $(k-1)^{n-1}$ proper path colorings in total. A coloring ending in the first color arises uniquely by appending that color to one of the B_{n-1} paths. Hence

$$B_n = (k-1)^{n-1} - B_{n-1}, \quad B_1 = 0.$$

Induction gives

$$B_n = \frac{(k-1)^n + (-1)^n(k-1)}{k}.$$

Indeed substitution into the recurrence gives this expression, and its value at $n = 1$ is 0. Closing the path into a cycle requires exactly that its endpoints differ. There are k choices for the first color, so the required number is $kB_n = (k-1)^n + (-1)^n(k-1)$. For $k = 0$ or 1, both the coloring count and the claimed expression are 0.

- (b) For C_6 with four colors the answer is $3^6 + 3 = 732$.

(c) The closing-edge correction has sign $(-1)^n$, so it adds $k-1$ for even n and subtracts $k-1$ for odd n . With two colors this distinction is especially clear: alternating colors close consistently on an even cycle (two colorings), but not on an odd cycle (zero colorings). $\square$

12. (The Pigeonhole Principle)

- (a) Prove that among any $n+1$ integers, there must be two whose difference is divisible by n .
- (b) Suppose 51 integers are chosen from

$$[100] = \{1, 2, 3, \dots, 100\}.$$

Prove that among the chosen integers, there must be two such that one divides the other.

Proof. (a) For $n \geq 1$, there are only n residue classes modulo n . Among $n+1$ integers, two have the same residue by the pigeonhole principle. Their difference is divisible by n .

(b) Every positive integer has a unique form $2^r q$, where q is odd and $r \geq 0$. For numbers at most 100, the odd part q belongs to the 50-element set $\{1, 3, 5, \dots, 99\}$. Among 51 distinct chosen integers, two therefore have the same odd part: $2^r q$ and $2^s q$, with, say, $r < s$. Since $2^s q = 2^{s-r}(2^r q)$, the smaller divides the larger. $\square$

13. (Recurrence Relations) Let a_n be the number of binary strings of length n that do not contain consecutive 1's.

- (a) Find the recurrence relation for a_n .
- (b) State the initial conditions.
- (c) Solve the recurrence or identify the sequence.
- (d) Compute a_6 and list all valid binary strings of length 4.

Solution. (a) For $n \geq 2$, a valid string either starts with 0, followed by any valid length- $(n-1)$ string, or starts with 10, followed by any valid length- $(n-2)$ string. Thus $a_n = a_{n-1} + a_{n-2}$.

(b) $a_0 = 1$ for the empty string and $a_1 = 2$ for the strings 0, 1.

(c) With $F_0 = 0, F_1 = 1$, we have $a_n = F_{n+2}$. Equivalently,

$$a_n = \frac{\phi^{n+2} - \psi^{n+2}}{\sqrt{5}}, \quad \phi = \frac{1 + \sqrt{5}}{2}, \quad \psi = \frac{1 - \sqrt{5}}{2}.$$

(d) The values a_2, a_3, a_4, a_5, a_6 are 3, 5, 8, 13, 21, so $a_6 = 21$. The eight valid length-4 strings are

0000, 0001, 0010, 0100, 0101, 1000, 1001, 1010.

$\square$

14. (Integer Solutions and Stars and Bars) Find the number of integer solutions to

$$x_1 + x_2 + x_3 + x_4 = 20$$

under each condition.

- (a) $x_i \geq 0$ for all i .
- (b) $x_i \geq 2$ for all i .
- (c) $x_1 \geq 3, x_2 \geq 0, x_3 \geq 5$, and $x_4 \geq 1$.
- (d) $0 \leq x_i \leq 8$ for all i .

Solution. (a) Stars and bars gives $\binom{20+4-1}{4-1} = \binom{23}{3} = 1771$.

(b) Set $y_i = x_i - 2 \geq 0$. Then $\sum y_i = 12$, giving $\binom{15}{3} = 455$.

(c) Subtract the lower bounds 3, 0, 5, 1. The four resulting nonnegative variables sum to 11, giving $\binom{14}{3} = 364$.

(d) Start with all nonnegative solutions and let A_i be the event $x_i \geq 9$. Specifying j such violations and subtracting 9 from each leaves a sum of $20 - 9j$. Three violations are

impossible, so inclusion–exclusion gives

$$\binom{23}{3} - \binom{4}{1}\binom{14}{3} + \binom{4}{2}\binom{5}{3} = 1771 - 1456 + 60 = 375.$$

□

15. (Set Partitions with Restricted Block Sizes) Let a_n be the number of ways to partition an n -element set into block sizes of 1, 2, or 3.

- (a) Write the exponential generating function for a_n .
- (b) Explain why the correct exponential generating function is

$$A(x) = \exp\left(x + \frac{x^2}{2!} + \frac{x^3}{3!}\right).$$

- (c) Find a summation formula for a_n .
- (d) Compute a_5 .

Solution. (a) The exponential generating function is

$$A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!} = \exp\left(x + \frac{x^2}{2!} + \frac{x^3}{3!}\right).$$

(b) A single allowed block has exponential generating function $x + x^2/2! + x^3/3!$. A partition is an unordered set of disjoint such blocks on labeled elements, so the exponential formula gives the claimed expression. Explicitly, expanding $\exp(x) \exp(x^2/2!) \exp(x^3/3!)$ independently chooses the numbers of singleton, pair, and triple blocks.

(c) If these numbers are b_1, b_2, b_3 , then

$$a_n = \sum_{\substack{b_1, b_2, b_3 \geq 0 \\ b_1 + 2b_2 + 3b_3 = n}} \frac{n!}{b_1! b_2! b_3! (2!)^{b_2} (3!)^{b_3}}.$$

The block factorials remove internal orders, and the $b_j!$ factors remove orders among blocks of equal size.

(d) For $n = 5$, the possible block-size patterns are $1 + 1 + 1 + 1 + 1$, $2 + 1 + 1 + 1$, $2 + 2 + 1$, $3 + 1 + 1$, and $3 + 2$. Their respective counts are 1, 10, 15, 10, 10, so $a_5 = 1 + 10 + 15 + 10 + 10 = 46$. □